

A Re:Dr (ours) – Dr.GRPO across three model scales Δ pass@8

Qwen 0.5B	+0.65	+0.19	+0.36	+0.28	+0.21	+0.34
Falcon 1B	+0.15	+0.31 [†]	+0.98	+0.64	+0.12	+0.44 [†]
Qwen 3B	+0.15	+0.14	+0.59	+0.41	+0.11	+0.28
	Graph	Countdown	Python	MathIR	Pantry	Average

 Δ PCMD

Qwen 0.5B	+0.56	+0.48	+0.08	+0.02	+0.33	+0.29
Falcon 1B	+0.22	+0.03 [†]	—	+0.02	+0.32	+0.19 [†]
Qwen 3B	+0.18	+0.10	+0.05	+0.01	+0.25	+0.13
	Graph	Countdown	Python	MathIR	Pantry	Average

B Initial reference + alternatives – terminal Dr.GRPO at Qwen2.5-0.5B Δ pass@8

Before training	+0.13	-0.38	-0.17	-0.29	+0.38	-0.07
Re:Dr (ours)	+0.65	+0.19	+0.36	+0.28	+0.21	+0.34
Re:Max (ours)	+0.62	+0.19	+0.51	+0.28	+0.20	+0.36
MaxRL (no replay)	+0.21	+0.10	+0.00	-0.07	+0.01	+0.05
UCPO	-0.00	+0.13	+0.00	+0.03	+0.05	+0.04
RLEP-Dr	+0.10	+0.10	+0.00	+0.13	+0.00	+0.07
Semantic-MaxEnt	+0.00	+0.14	+0.00	-0.12	+0.20	+0.04
GRPO	+0.04	+0.16	+0.00	-0.05	+0.02	+0.03
GAPO	+0.17	+0.01	+0.01	-0.05	+0.28	+0.08
SetPO	+0.06	-0.05	+0.50	+0.05	+0.15	+0.14
	Graph	Countdown	Python	MathIR	Pantry	Average

 Δ PCMD

Before training	+0.11	—	—	+0.03	+0.59	+0.24
Re:Dr (ours)	+0.56	+0.48	+0.08	+0.02	+0.33	+0.29
Re:Max (ours)	+0.53	+0.50	+0.25	+0.01	+0.32	+0.32
MaxRL (no replay)	+0.09	+0.01	+0.00	-0.00	+0.00	+0.02
UCPO	-0.00	+0.02	+0.00	-0.00	+0.00	+0.01
RLEP-Dr	+0.03	+0.01	+0.00	-0.00	+0.00	+0.01
Semantic-MaxEnt	+0.00	+0.02	+0.00	+0.00	—	+0.01
GRPO	+0.01	+0.00	+0.00	+0.00	+0.00	+0.00
GAPO	+0.13	+0.34	+0.00	+0.00	+0.65	+0.22
SetPO	+0.04	+0.01	+0.01	+0.00	+0.27	+0.07
	Graph	Countdown	Python	MathIR	Pantry	Average